Teaching Statement

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Teaching mathematics has been a source of joy for me since I began tutoring my siblings and family friends in middle school. I went on to teach high school algebra and later became a TA at UCLA, and my enthusiasm for teaching has grown with each experience. I want my students to ask questions freely, develop confidence in their own reasoning, and discover something in mathematics that they want to pursue further.

One of my guiding principles is to **provide two answers**. Students learn in different ways and have different intuitions. Many advances in mathematics come from bringing together different perspectives, and I want students to appreciate that value in their own work. In my discussions, I frequently give two proofs of a theorem or two ways to solve a problem. For example, when teaching induction, I explain why the first n odd numbers sum to n^2 both algebraically and pictorially, by drawing a square built from successive L-shaped layers. The picture helps students see why the algebra works. A student in my fall 2023 Discrete Structures section described this approach.

> " *When you ask the same question again, he purposely puts in the time to explain the solution in a different way.* "

I also believe that **teaching should respond to students' interests and needs**. In 2021, I returned home as a City Year AmeriCorps team leader at Miami Jackson Senior High School. I taught four Algebra I classes, often in Spanish with students who had recently immigrated, and planned weekly support for eleven students. A particular challenge revealed itself in the first week when many told me math was their least favorite subject. To help change that perception, I connected algebra to their interests in live streaming, sports, and other media. After a few months, I was happy to hear some students say they looked forward to class. At UCLA, I connect course material to historical facts and current events. I also give students a say in planning discussions by asking what they find confusing and want me to explain further. A student in my winter 2023 Discrete Structures section described the effect.

> " *He often asked for the students' input on what we were confused about, which helped us focus on relevant material.* "

A third principle is to **give students time to find an answer of their own**. Many students come to UCLA used to understanding high school mathematics quickly. In college, they are learning to work through the discomfort of being confused. Artificial intelligence can reinforce the expectation of an immediate answer. I want students to feel comfortable spending time with a problem before I offer guidance. A student in my fall 2023 Multivariable Differential Calculus section appreciated this.

> " *He lets us think through the problems ourselves before providing guidance, which I wholly appreciate.* "

A student in another calculus course, however, explained that when they felt completely lost, they wanted a worked example. These responses highlight the judgment involved in deciding when to help. I ask students to verbalize their thinking, even if they have written nothing down. This helps me hear whether they need a starting point or more time with an idea. It can also help them recognize that they are already on the right track.

My experience as UCLA's Math Learning Assistant (LA) Coordinator in 2024–25 also helped me grow as an instructor. I led discussions about pedagogy with undergraduates who supported students in small groups, helped LAs with course mathematics, and worked with instructors to expand the program. I want an LA approaching a table to listen long enough to hear what idea the group is pursuing. Sometimes the person best placed to explain the next step is already sitting at the table. Discussing these choices with LAs has also helped me use active learning more effectively in my own sections.

Finally, I try to **share the joy I find in mathematics**. I want my classroom to feel friendly and supportive, so students feel comfortable asking questions, sharing ideas, and seeing the beauty in mathematics. Arthur Benjamin's Discrete Mathematics course, which I took in my second year at Harvey Mudd College, helped inspire me to become a mathematician and study combinatorics. I still remember his delight in the subject and the number tricks he taught us using divisibility by nine. I try to bring that spirit to discussions by sharing interesting facts students can explain using what they have just learned. A student in my spring 2025 Enumerative Combinatorics section wrote about their growing confidence and enjoyment.

> " *Before the class, I was very uncertain about my skills in combinatorics. However, seeing Thomas every week made me gain confidence week by week and now I actually enjoy combinatorics, something I did not know was possible.* "

I try to share this joy beyond discussions as well. Through UCLA's Directed Reading Program, I have mentored nine students in the probabilistic method, graph theory, and combinatorial proof. In 2025–26, Olha Shevchenko and I co-mentored three undergraduates on q-Fibonomial coefficients. After a quarter introducing them to combinatorial proofs involving q-Fibonacci numbers, we began studying a unimodality conjecture of Nantel Bergeron, Cesar Ceballos, and Josef Küstner. All three students are coauthors on our preprint proving it in several cases.

Teaching has been central to my mathematical life, and I look forward to continuing to grow as an instructor by learning from my students and colleagues. In every course I teach, I want to offer clear explanations, make room for students ' ideas, and build a classroom where they feel supported as they work through difficult concepts. Helping students become more confident and independent while sharing my enthusiasm for mathematics is what makes teaching so rewarding to me.